

Topology General Exam

August 26, 2017

Name:

Instructions: This is a four hour exam and 'closed book'. There are eight problems. Show your work using methods and results from the first year topology course topics. Results from one part of a problem can be assumed in later parts.

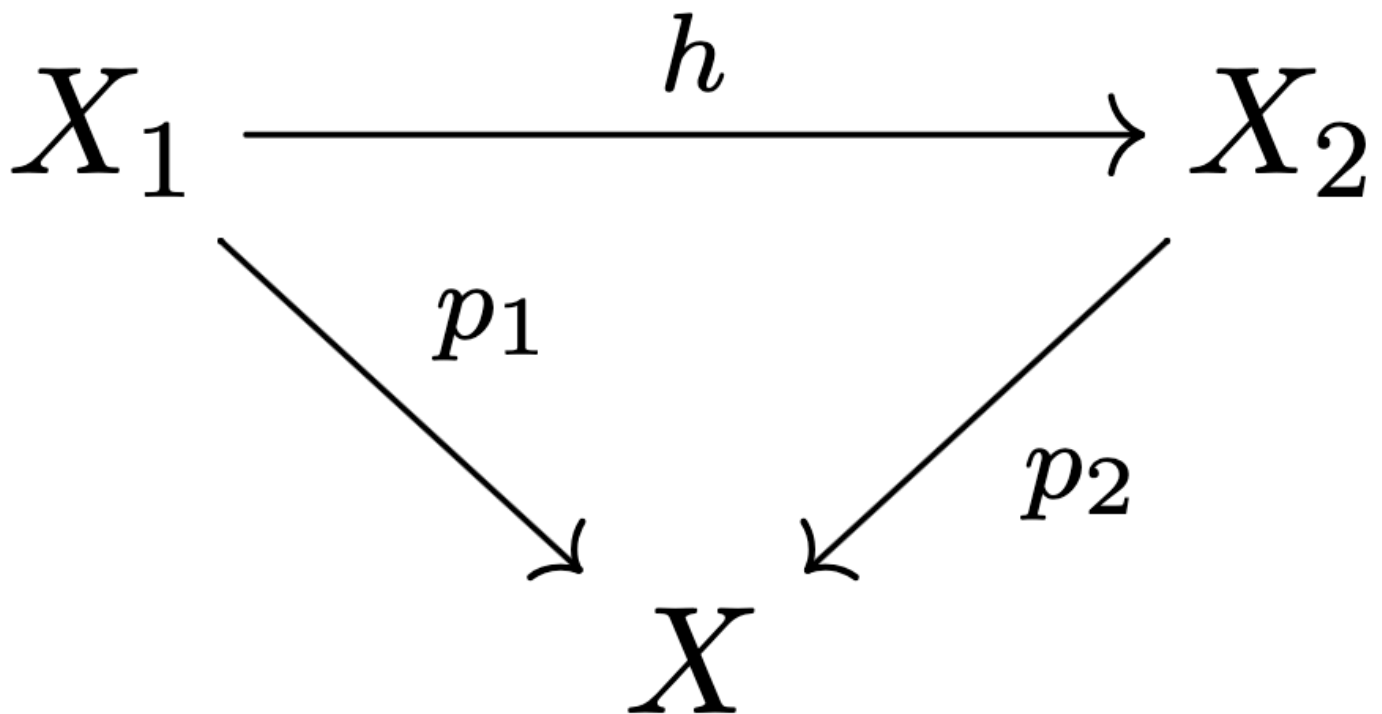
Problem 1

Let $f : \mathbb{R}^2 - \{(0, 0)\} \rightarrow \mathbb{R}^3$ be given by $f(x, y) = (x^2, y^2, xy)$.

- (a) Prove that f is an immersion, but not an embedding.
- (b) Find all values of c such that f is transverse to the plane $\{(u, v, w) \mid v = c\} \subset \mathbb{R}^3$.

Problem 2

Two covering spaces over X , $p_1 : X_1 \rightarrow X$ and $p_2 : X_2 \rightarrow X$, are said to be isomorphic if there exists a homeomorphism $h : X_1 \rightarrow X_2$ such that the diagram commutes:



The triangle has h from X_1 to X_2 along its top edge, p_1 from X_1 to X , and p_2 from X_2 to X . Both p arrows point down to X . Commutativity means p_2 composed with h equals p_1 .

Describe the isomorphism classes of covers of the space $\mathbb{R}P^2 \times \mathbb{R}P^2$.

Problem 3

Suppose that $t \in \mathbb{R}$ is a regular value of a smooth map $f : \mathbb{R}^n \rightarrow \mathbb{R}$, and let $M = f^{-1}(t)$.

- (a) Will M necessarily have a nowhere vanishing normal vector field?
- (b) Will M necessarily have a nowhere vanishing tangent vector field?
- (c) Will M necessarily be orientable?

In each part, explain how to construct the vector field or orientation if it must exist, and give a counterexample, if it needn't.

Problem 4

- (a) Define what it means for two chain maps between chain complexes to be chain homotopic. Then prove that if $f_*, g_* : C_* \rightarrow D_*$ are chain homotopic, then $f_* = g_* : H_*(C_*) \rightarrow H_*(D_*)$.
- (b) Call a chain map a quasi-isomorphism if it induces an isomorphism on homology. Let

$$\begin{array}{ccccccc}
0 & \longrightarrow & A_* & \longrightarrow & B_* & \longrightarrow & C_* \longrightarrow 0 \\
& & \downarrow f_* & & \downarrow g_* & & \downarrow h_* \\
0 & \longrightarrow & D_* & \longrightarrow & E_* & \longrightarrow & F_* \longrightarrow 0
\end{array}$$

The upper short exact sequence of chain complexes is $0 \rightarrow A_* \rightarrow B_* \rightarrow C_* \rightarrow 0$. The lower sequence is $0 \rightarrow D_* \rightarrow E_* \rightarrow F_* \rightarrow 0$. Downward chain maps are f_* from A_* to D_* , g_* from B_* to E_* , and h_* from C_* to F_* . Both squares commute.

be a commutative diagram of chain complexes, such that each horizontal row is exact. Show that if f_* and g_* is a quasi-isomorphism then so is h_* .

Problem 5

Let $f : M \rightarrow N$ be a smooth map between manifolds of dimension m and n respectively. Let $D(f) \subset M \times M$ be the 'double point' subspace:

$$D(f) = \{(x, y) \mid x \neq y \text{ and } f(x) = f(y)\}$$

(a) Say that f is self transverse if for all $(x, y) \in D(f)$, $d_x f(T_x M) + d_y f(T_y M) = T_{f(x)} N$. Show that then $D(f)$ is a smooth manifold of $M \times M$, and find its dimension. [Hint: Show that if f is self transverse, then the function $f \times f : M \times M - \Delta(M) \rightarrow N \times N$ is transverse to the diagonal $\Delta(N) \subset N \times N$.]

(b) Describe, with pictures and/or words, an example of a self transverse smooth function $f : S^1 \rightarrow \mathbb{R}^2$ for which $D(f)$ is nonempty.

(c) Describe, with pictures and/or words, an example of a smooth function $f : S^1 \rightarrow \mathbb{R}^2$ that is not self transverse.

Problem 6

(a) Show that there is no continuous map $g : S^2 \rightarrow S^2$ such that, for all $x \in S^2$, $g(x) \neq x$ and $g(x) \neq -x$. [Hint: if such a g exists, explain how it can be used to show that the antipodal map is homotopic to the identity map on S^2 . Hmm ...]

(b) Show that every continuous map $f : \mathbb{R}P^2 \rightarrow \mathbb{R}P^2$ has a fixed point. [If f has no fixed point, use covering space theory to show that one can construct g as in part (a).]

Problem 7

Suppose a finite group G acts freely on the right of a Hausdorff space X . Let X/G be the space of G -orbits - sets of the form $\{xg \mid g \in G\}$ - given the quotient topology.

(a) Show that the quotient map $X \rightarrow X/G$ is a covering space map. [Hint: you need to show that each point $x \in X$ has an open neighborhood U such that all the translates of U — the sets Ug for $g \in G$ - are disjoint.]

(b) How are the groups G , $\pi_1(X)$ and $\pi_1(X/G)$ related?

(c) Show that there exists a smooth 3-manifold with a fundamental group that is both finite and non-abelian. [One approach: Recall that S^3 can be viewed as the quaternions of unit length.]

Problem 8

(a) S^2 has a CW complex structure with one 0-cell and one 2-cell, and then the associated 'product' CW structure on $S^2 \times S^2$ has one 0-cell, two 2-cells, and one 4-cell. Compute the homology groups of $S^2 \times S^2$.

(b) If M is a smooth connected n -dimensional manifold, let $\widetilde{M}$ denote M with a small open n -ball removed. Show that $\widetilde{S^2 \times S^2}$ is homotopy equivalent to $S^2 \vee S^2$. [Hint: you can assume that the small open 4-ball is removed from the interior of the 4-cell.]

(c) The connected sum $M \# N$ of two n -manifolds admits a decomposition $M \# N = \widetilde{M} \cup \widetilde{N}$ with $\overline{\widetilde{M}} \cap \overline{\widetilde{N}}$ diffeomorphic to S^{n-1} . Compute the homology groups of $(S^2 \times S^2) \# (S^2 \times S^2)$.